

QVerse — Week 12: Quantum Fourier Transform — Useful Intuition

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

2026

How this booklet fits the course

This theory booklet is designed to be read alongside `12_quantum_fourier_transform.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 12; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Describe QFT as a basis transformation acting on amplitudes.
- Use `QFTGate` and its inverse.
- Verify QFT followed by inverse QFT.
- Inspect decomposition and scaling.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. Classical Fourier idea

A classical Fourier transform changes representation. For example, a time-domain signal can be represented by its frequency components.

The Quantum Fourier Transform (QFT) is not simply “a faster FFT.” It is a unitary basis transformation acting on the amplitudes of a quantum state.

For

$$N = 2^n,$$

the QFT is defined on computational-basis states by

$$|j\rangle \mapsto \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j k / N} |k\rangle.$$

By linearity, this definition determines its action on every n -qubit state.

2. What the QFT does conceptually

For a basis state $|j\rangle$, the QFT produces a superposition with equal amplitude magnitudes but a structured pattern of phases:

$$\frac{1}{\sqrt{N}} (|0\rangle + e^{2\pi i j / N} |1\rangle + e^{2\pi i 2j / N} |2\rangle + \dots).$$

Therefore a computational-basis label is converted into a phase pattern.

This is why measuring immediately after QFT usually loses the interesting information: all basis states may have equal probabilities even though their phases differ.

The QFT becomes useful when those phases are later interfered with or converted into measurable binary information.

3. Matrix form

The QFT matrix has entries

$$F_{jk} = \frac{1}{\sqrt{N}} e^{2\pi i jk/N}.$$

It is unitary:

$$F^\dagger F = I.$$

Therefore its inverse is

$$F^{-1} = F^\dagger.$$

The inverse QFT reverses the phase-encoding transformation.

4. Small example: one qubit

For $n = 1$,

$$N = 2.$$

The QFT matrix becomes

$$F_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H.$$

So the one-qubit QFT is simply the Hadamard gate.

This connects the QFT to the interference ideas introduced much earlier.

5. Binary-fraction interpretation

For multi-qubit states, QFT can be decomposed into single-qubit Hadamards and controlled phase rotations.

If a basis label has binary expansion

$$j = j_{n-1} \dots j_1 j_0,$$

the transformed state can be factored into qubits whose phases contain binary fractions of j .

One common expression is

$$\text{QFT}|j\rangle = \frac{1}{2^{n/2}} \bigotimes_{m=1}^n (|0\rangle + e^{2\pi i 0.j_m j_{m+1} \dots j_n} |1\rangle),$$

up to qubit-order conventions.

The exact indexing is less important for this course than the structural idea: controlled phase gates build the required phase relationships bit by bit.

6. Circuit decomposition

A standard QFT circuit uses:

- Hadamard gates;
- controlled phase rotations;
- optional final swaps to reverse qubit order.

The controlled rotations have decreasing angles such as

$$\frac{\pi}{2}, \frac{\pi}{4}, \frac{\pi}{8}, \dots$$

as qubits become farther apart in significance.

The final SWAP network corrects the natural reversed output order of the decomposition.

7. Exact scaling

The standard exact QFT decomposition uses $O(n^2)$ two-qubit controlled-phase operations.

The quantum circuit is therefore polynomial in the number of qubits n , even though the corresponding $N \times N$ Fourier matrix has dimension

$$N = 2^n.$$

This compact decomposition is one reason QFT is an important quantum primitive.

However, efficient circuit implementation does not mean we can simply read all N Fourier coefficients. Measurement still produces limited classical information.

8. Approximate QFT

Small-angle controlled rotations contribute less to the final state and can sometimes be omitted.

An approximate QFT can reduce depth while maintaining sufficient accuracy for a larger algorithm.

This trade-off matters in realistic hardware where long circuits are more error-prone.

The Week 12 notebook uses the exact library gate for clarity but inspecting the decomposition helps students see where the cost comes from.

9. QFT followed by inverse QFT

Because the transform is unitary,

$$F^{-1}F = I.$$

Therefore for any state

$$|\psi\rangle,$$

$$\text{QFT}^{-1}\text{QFT}|\psi\rangle = |\psi\rangle.$$

Testing this round trip on all 3-qubit basis states is a good software validation exercise.

A stronger check compares the full circuit operator with the identity up to global phase.

10. Why QFT is usually a subroutine

QFT is central to:

- quantum phase estimation;

- period finding;
- Shor's factoring algorithm;
- some arithmetic and signal-processing-style routines.

The QFT itself does not usually produce a useful answer if immediately measured. It changes representation so that another algorithmic step can extract phase or periodicity information.

This is analogous to using a classical Fourier transform as one stage in a signal-processing pipeline.

11. Connection to phase estimation

Suppose an eigenstate satisfies

$$U|\psi\rangle = e^{2\pi i\varphi}|\psi\rangle.$$

Quantum phase estimation encodes multiples of the phase φ across a register and then applies the **inverse QFT** to convert that phase pattern into an approximate binary representation of φ .

This is one of the most important reasons to understand QFT.

The course notebook does not implement full QPE, but the conceptual link should be clear.

12. Measurement and lost phase information

If QFT maps a basis state to equal-magnitude amplitudes, immediate computational-basis measurement may appear uniform.

A uniform histogram does **not** mean the QFT “did nothing.” The input value is encoded in relative phases.

This is another example of the principle from Week 4:

$$\text{phase information} \neq \text{direct measurement probability}.$$

Useful circuits must convert that phase structure into observable amplitude differences.

13. Bit ordering and swaps

QFT circuits are especially sensitive to qubit ordering.

The theoretical QFT may naturally produce output qubits in reversed significance order. Libraries may include swaps, expose a `do_swaps` option, or present the decomposition in a convention that requires careful interpretation.

The safest practice is to:

1. state the convention;
2. test a small basis state;
3. verify QFT + inverse QFT;
4. avoid assuming that visual wire order equals binary significance.

14. Common misconceptions

“QFT is the same algorithm as classical FFT”

They are mathematically related transforms but have very different input/output access models.

“QFT outputs the Fourier spectrum as a classical array”

No. It outputs a quantum state.

“A uniform histogram means no information is present”

The relevant information may be in relative phases.

“QFT alone is Shor’s algorithm”

QFT is one important subroutine inside a larger period-finding workflow.

15. What to know before the notebook

You should understand the definition

$$|j\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i jk/2^n} |k\rangle,$$

why the QFT is unitary, why the inverse recovers the input, and why the useful information often resides in phases rather than immediate measurement probabilities.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.